

• Medium → Es # 11 → Container with Most Water

▲ Problem Statement → Given an integer array "height", find two lines that form a container capable of holding the maximum amount of water.

• The container can't be slanted

▲ Main Problem Solution → Use the "Two Pointer" approach.

- Start with one pointer at the beginning & one at the end.
- The area is calculated as:- " $\text{width} \times \text{minimum height}$ "

Move the pointer with the smaller height, because making the taller one can't increase the container's height.

▲ Brute Force Method • Try every possible pair of lines & calculate the area.

Time $\rightarrow O(n^2)$

Space $\rightarrow O(1)$

This is too slow for " $n = 10^5$ "

▲ Algorithm • 1. Set " $\text{left} = 0$ " & " $\text{right} = n - 1$ "

2. Calculate the current area:-

$$\text{min}(\text{height}[\text{left}], \text{height}[\text{right}]) \times (\text{right} - \text{left})$$

3. Update the maximum area.

4. If " $\text{height}[\text{left}] < \text{height}[\text{right}]$ ", move " left " forward.

5. Otherwise, move " right " backward.

6. Repeat until " $\text{left} \geq \text{right}$ "

7. Return the maximum area.

▲ Important Points → • Use two pointers starting from both ends.

• Area = " $\text{width} \times \min(\text{height}[\text{left}], \text{height}[\text{right}])$ "

• Always move the pointer with the smaller height.

• Width decreases with every move, so we need a chance to increase the height.

• " maxWater " stores the maximum area found.

▲ Complexity → Time :- $O(n)$
Space :- $O(1)$